The Energy Cost of Prompting a Large Language Model

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Abstract

This study aims to quantify the energy demands of large language models through the perspective of a casual user. First, the authors introduce a dataset of 1,000 prompts annotated by length, task type and complexity that is created to be representative of an average LLM prompt. The authors report both the experimental energy cost by running the prompts through an LLM, and the simulated energy cost by calculating the necessary floating-point operations for each prompt. Experimental results are recorded in Watts per hour and are collected by measuring GPU and CPU usage in Watts every 0.5 seconds, then averaging the measurements and calculating overall Watt-hours with respect to model response time. Simulated results are in Joules.

The authors found that the average energy cost of a prompt is approximately 0.1 Watt-hours for Facebook's opus-125M model and 0.09 for OpenAI's GPT-2-124M model, which aligns with existing articles in which Google reported an average cost of 0.24 Watt-hours per prompt for their Gemini model [4], and a blog post in which OpenAI CEO Sam Altman states ChatgGPT uses 0.34 Wh per prompt [2].

The findings in this study also show the length of LLM output has a significant impact on energy cost. Similarly, the length of LLM input significantly impacts energy cost. As expected, long and complex prompts cost more energy than shorter, less complex prompts.

The dataset and source code for the experiment are available at github.com/abbyPitcairn/llmenergy.

CCS Concepts

• **Sustainability**; • **Large Language Models**;

Keywords

AI, LLM, NLP, Energy, Green AI, Sustainability

ACM Reference Format:
Abigail Pitcairn. 2026. The Energy Cost of Prompting a Large Language Model. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 8 pages. https://doi.org/XXXXXXX.XXXXXXX

ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/XXXXXXX.XXXXXXX

Table 1: Energy consumption (in Watt-hours) for common activities and a single GPT-2-124M prompt inference.

Activity	Energy
One GPT-2-124M prompt	**0.09 Wh**
One Google search in 2009 [14]	0.30 Wh
LED bulb (60 W) for 1 minute	1.00 Wh
LED TV, 40–50 in. (60 W) for 0.5 hrs	30.00 Wh

1 Introduction

The field of artificial intelligence (AI) has been growing rapidly during recent years. We've seen more advertisements for AI products and tools, we talk more about AI use in our education, and casual AI use is becoming more widespread [19].

One AI model rising in popularity is OpenAI's ChatGPT. ChatGPT, based on their GPT model, is a transformer-based large language model (LLM). LLMs have gained popularity due to their conversational ability, with ChatGPT being promoted as something you can talk to and use as an everyday tool [13].

The normalization of casual AI use has come with criticisms. Recent studies raise concerns about the impact on cognitive function [9], public health [1], and the environment. The amount of power needed to train and run AI models and the impact of this energy demand on natural resources, clean water supplies and air quality, is especially of concern.

Lots of existing work has gone into measuring large-scale energy costs associated with AI, such as training LLMs [20], and the fine-scale energy costs, such as the cost per token during inference [15]. We focus our work on the consumer side: our goal for this research is to inform personal decisions about LLM usage in the context of reducing one's own environmental impact.

In Table 2, we compare our findings for LLM energy cost to common activities, such as powering a light bulb or searching on Google.

2 Proposed Approach

Our goal is to investigate the following two research questions.

RQ1: What is the average energy cost of issuing a prompt and receiving a response from a modern LLM?

RQ2: What are some qualities of prompts that make them more or less computationally expensive?

We hypothesize that our experimental and simulated results will match the existing literature and that we will be able to report an accurate measurement of energy cost per LLM prompt, and we expect to see positive correlation between input and output length and energy cost.

Our approach has two steps: first, we gather experimental results by running our dataset through popular LLMs from HuggingFace[1]. We query the hardware system for CPU and GPU energy consumption in Watts every 0.5 seconds during inference and record the model's response time per prompt. We then calculate energy cost in Watt-hours by taking the product of the average of our Watt measurements and the response time converted to hours.

2.1 Dataset

We created our own dataset for this study. The dataset is a combination of LLM-generated text prompts and prompts from five popular datasets from HuggingFace filtered by *question answering* and *code generation* task types. To create the dataset, first we prompted OpenAI's ChatGPT[2] 4.0 model to generate 500 unique prompts. We stored these prompts to a file, which is available on our Github. Then we took 100 rows from each of the five HuggingFace datasets using a random seed and setting a maximum token count of 250. Prompts over 250 tokens were skipped and not added to our dataset.

When the dataset generation script is run, it combines the LLM-generated prompts with 500 HuggingFace prompts taken directly from the website. Our code uses HuggingFace tokens for verification. The final step was verifying the combined dataset and annotating its subjective qualities by hand.

Below are the online datasets we used with URLs and descriptions:

- **Math-500**[3]: a subset of 500 math questions from OpenAI's math-question dataset released in their paper, *Let's Verify Step by Step* [10].
- **MMLU**[4]: multiple-choice questions from a range of core school subjects.
- **Human Eval**[5]: 164 handwritten coding problems with function signatures, doc strings, and examples of input and output, also published by OpenAI [5].
- **Opus Reasoning**[6]: includes a problem, thinking, and solution. In this study we use only the problem statement as our prompt.
- **Simple QA**[7]: question and answer pairs created to test LLM factuality.

Each prompt in our dataset has three labels for our analysis.

- **Prompt length:** measured by the number of words, also called tokens. Additionally we sort prompts into four buckets based on length so that 25% of prompts fall into each length category.
- **Task type:** we created a key for task types so they can be simply annotated as one of the following task type categories:
  - 1 = Casual / small talk
  - 2 = Explanation / learning
  - 3 = Writing / rewriting
  - 4 = Coding / technical help
  - 5 = Productivity / planning / structured
  - 6 = Creative / storytelling
- **Task complexity:** this label is closely tied to task type and prompt length, but they are not the same. Complexity is defined as:
  - Low (0): a single thought or question that does not require reasoning, i.e. "Hi", "Explain why the ocean is salty", etc.
  - Medium (1): more than a single sentence, or requiring reasoning.
  - High (2): a prompt that is more than a paragraph and requires many layers of reasoning, i.e., complex code generation, multi-step math, etc.
- **Origin:** this label says where the prompt originated from, which is recorded as either "ChatGPT" or the abbreviated name of the HuggingFace dataset.

As the only annotator was the author, we acknowledge that task type and complexity labels are subject to bias. However, these labels are not seen by the models and are solely for analysis purposes.

In this research, we do not record model output. This is left for future work.

Table 2: Examples of Prompts from ChatGPT with Annotations.

Prompt	Length	Task	Complexity
Suggest a creative weekend activity.	5	1	0
Explain what molecules are.	4	2	1
Improve: 'The data are confusing.'	5	3	0

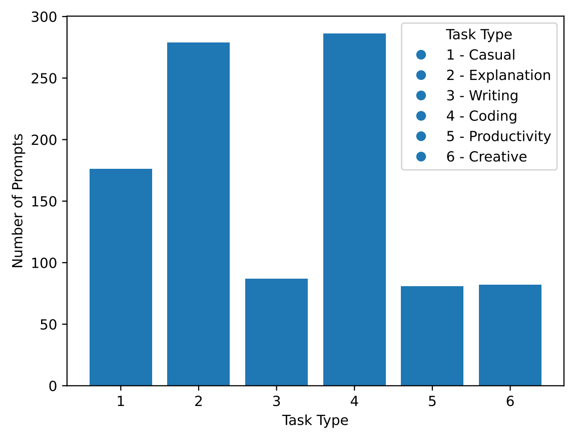

Figure 1: Distribution of Task Types.

[1] https://HuggingFace.com
[2] https://ChatGPT.com
[3] https://huggingface.co/datasets/HuggingFaceH4/MATH-500
[4] https://huggingface.co/datasets/cais/mmlu
[5] https://huggingface.co/datasets/openai_humaneval
[6] https://huggingface.co/datasets/Opus-4.6-Reasoning-3300x
[7] https://huggingface.co/datasets/simpleqa-verified

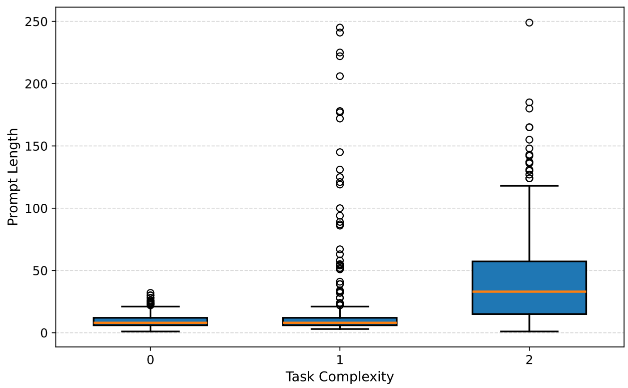

Figure 2: Distribution of Prompt Lengths by Complexity.

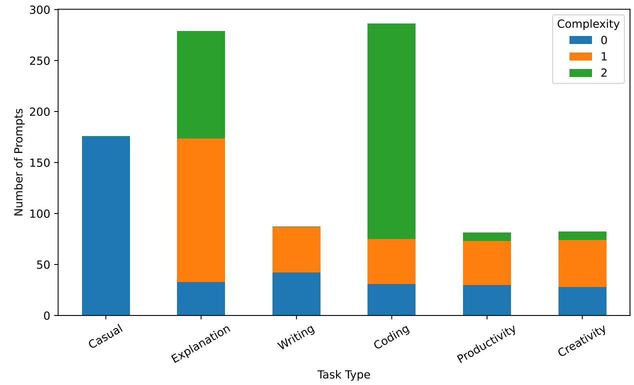

Figure 4: Distribution of Prompt Complexity by Task Type.

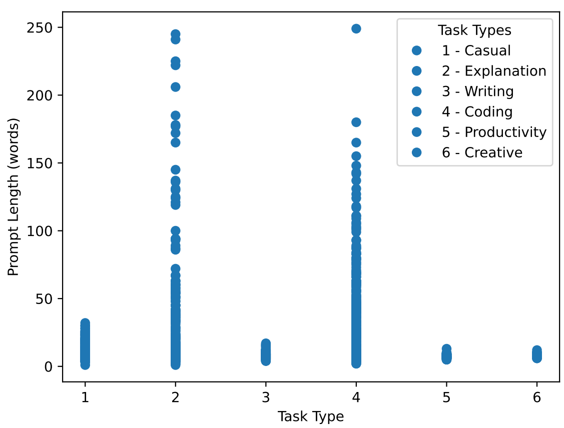

Figure 3: Distribution of Prompt Lengths by Task Type.

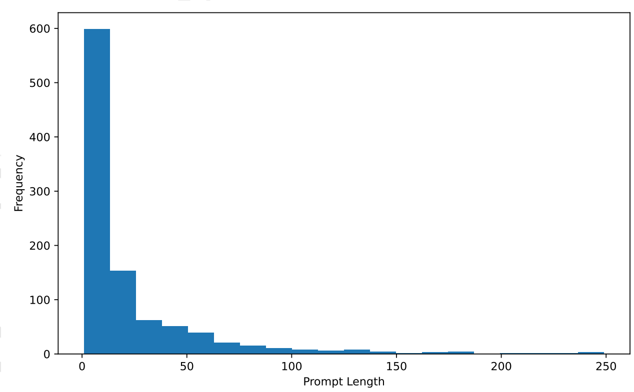

Figure 5: Distribution of Prompts by Token Count.

2.2 Data Visualization

Fig. 1 shows the frequency of each task type in the dataset. Prompts are concentrated in Explanation and Coding, reflecting a computer science student's usage patterns: we assume a correlation between awareness of AI tools and interest in coding-related tasks, and LLMs are actively encouraged in educational settings and advertisements to be used for explanation tasks. Casual prompts are moderately represented, as online LLM interfaces typically give suggestions for casual prompts on their welcome screens. Writing, Productivity, and Creativity appear least frequently.

Fig. 2 shows prompt length distributions by complexity label. As expected, low complexity prompts (*0*) are all under 50 words. Medium complexity prompts (*1*) range from 0-250 words with a median near 25, and high complexity prompts (*2*) are generally the longest. Fig. 2 implies a positive correlation between complexity and prompt length.

Fig. 3 shows prompt lengths by task type. Consistent with Fig. 1, Explanation and Coding (types 2 and 4) dominate in both the number of representative prompts and the range of lengths.

Fig. 4 compares complexity distributions across task types. Explanation and Coding show the most balanced complexity distributions and together account for the majority of high-complexity prompts. Casual (type 1) contains only low-complexity prompts, as expected: small-talk requires no multi-step reasoning. Writing (type 3) splits roughly evenly between low and medium complexity. Productivity and Creativity (types 5 and 6) show similar distributions, predominantly low and medium complexity, likely due to the similar nature of the tasks.

Fig. 5 shows the overall prompt length distribution across the dataset. Lengths follow a long-tail distribution, with most prompts under 50 words and sparse representation beyond 100 words. We assume this to be reflective of real-world usage where casual users tend to favor short, direct prompts.

Fig. 6 visualizes the most frequent terms across all prompts. Prominent terms relate to coding tasks, while *return* also appears frequently in information retrieval contexts, where users ask the LLM to return a specific piece of information.

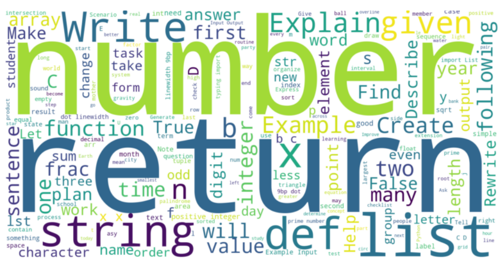

Figure 6: Frequency of Terms in Prompt Text.

2.3 Models

The models used were chosen from HuggingFace's most-downloaded page, so that results are relevant to models people are using currently. Additionally, we only selected models with the text-to-text generation tag to ensure they were fit for this task. Due to limited memory space in our hardware setup, we selected only lightweight models for our experiment.

Model parameters are not changed from the default settings, except for the parameters described in A. Models used include:

- **GPT-2:**[8] this generative pre-trained transformer architecture is a popular model for text generation. We use the 125-million parameter version, first released by OpenAI in 2019 [16]. Following Chien et al., we chose a GPT model due to its rising popularity [6].

- **Facebook-opt:**[9] this lightweight model (124-million parameters) was released by Meta AI in 2022 and recieved 6.7 million downloads off HuggingFace in March, 2026 [21].

2.4 Hardware

For hardware, a MacBook Air was used to securely connect and run the experiment scripts remotely on a Linux server running Ubuntu 22.04.5 LTS (kernel 6.8.0-48-generic). The system is equipped with a single Intel Xeon Gold 5215 processor (10 cores, 20 threads, 2.50 GHz base / 3.40 GHz boost), 64 GB of DDR4 RAM, and two NVIDIA GeForce RTX 2080 Ti GPUs (11 GB GDDR6 each). Primary storage consists of a 1.9 TB NVMe SSD (Toshiba XG5), supplemented by a 16.4 TB HDD and network-attached storage totaling 44 TB. The authors would also like to acknowledge that the server is shared among other university staff and students, and therefore other processes running on the machine could not be fully controlled for.

3 Results

3.1 Experimental Results

These results establish an average energy cost of **0.087 Wh** per GPT-2 prompt and the median energy cost was **0.114 Wh**. Fig. 7 shows the distribution of prompts by energy cost. Energy varies by a factor of 30 across all prompts. The cheapest prompts consumed just 0.005 Wh (18 joules) and completed in 0.1 seconds, while the most expensive prompt reached 0.154 Wh (555 joules) and took 2.7 seconds.

[8]https://huggingface.co/openai-community/gpt2
[9]https://huggingface.co/facebook/opt-125m

We identified that a significant determiner of energy cost is the number of output tokens the model generates. Recorded Watt-hours per prompt differ significantly by output length, with a Mann-Whitney U rank test showing significant difference in energy cost for prompts that produced over 100 output tokens versus prompts producing under 100 tokens ($p < 0.05$). Prompts producing over versus under 100 output tokens had about 800 and 200 samples respectively.

Fig. 10 shows the distribution of output length measured in tokens for each data origin. We capped output length at 250 tokens, and the outputs reached that limit about 60% of the time. The distributions look very similar to Fig. 9, further confirming the relationship between energy cost and output length.

Even though we see output length cluster around exactly 250 tokens as evident in Fig. 11, energy cost has more variance. It's possible this is due to there not being a cap on energy consumption like there is on output tokens, but it also implies other factors are determining total energy cost.

3.2 Simulated Results

The Joules calculated from simulated results support our experimental measurements.

Fig 12 shows that high-complexity coding tasks cost the most in terms of Joules per input token, while high-complexity explanation tasks come in as a distant second.

4 Weaknesses

There are a couple of weaknesses we are aware of in the research methods that we would like to address.

- **Inflated Energy Cost on Prompt 1**: The very first prompt in the experiment (run 1, prompt 1) had a reported energy cost that was not accurate because it did not account for the energy cost of starting the model. After this prompt, there are no similar outliers. However, we did not identify this error until after results analysis. The correct value for average combined CPU/GPU for prompt ID#1 was 0.0975 Wh, while for the first run prompt ID#1 was closer to 0.115 (+0.0175).
- **Non-Standard Online Datasets**: Some of our prompts in the dataset are pulled from online datasets, and not all of these online datasets are perfectly cleaned. In particular, within the Opus Reasoning subset, we had two outliers where the problem statements were *"# Problem"* and *". What language is this?"*. Additionally, the reproducibility of our experiment does not account for any changes that may be made to the online HuggingFace datasets.
- **Bias in the Dataset**: In our attempt at creating a dataset representative of the average LLM-prompt, we acknowledge that some of the author's personal biases influenced which HuggingFace datasets were chosen. For example, there is an emphasis on coding-related prompts that may be biased due to the author's occupation.

5 Related Work

The existing body of research on LLM energy consumption addresses both the training and inference phases across a range of

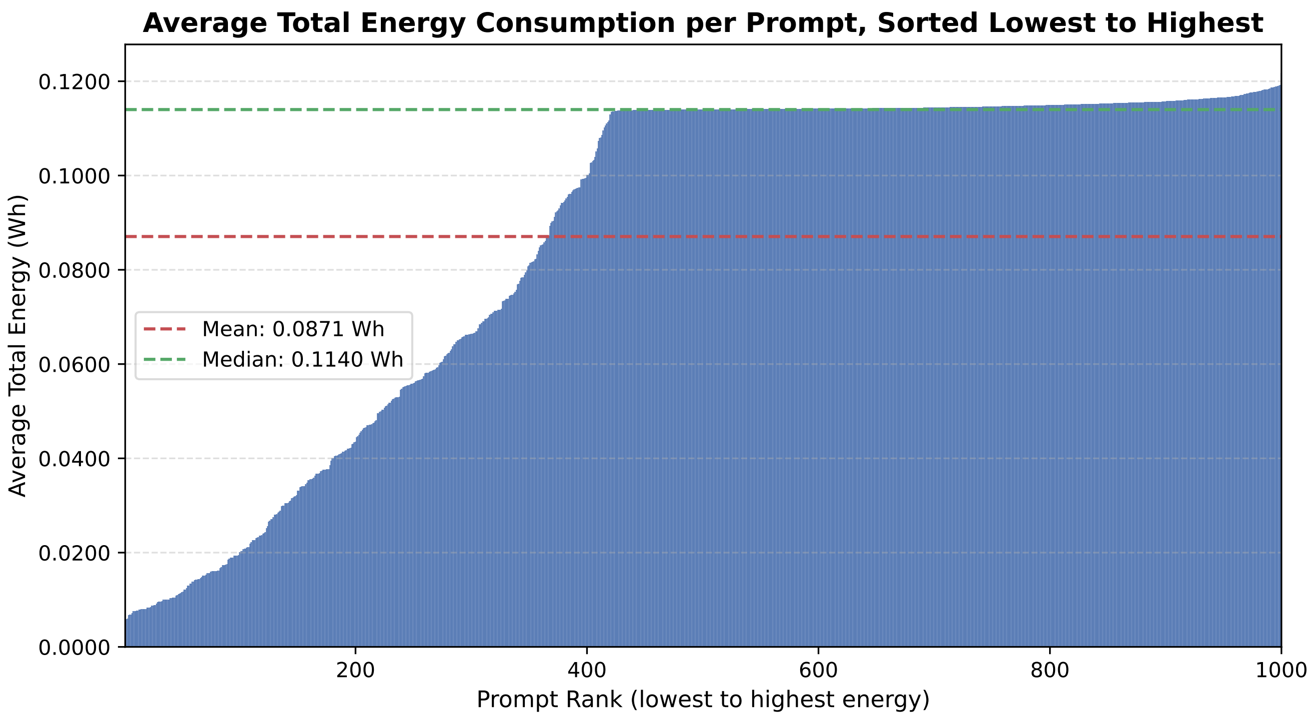

Figure 7: Distribution of Energy Cost per Prompt in Watt-Hours.

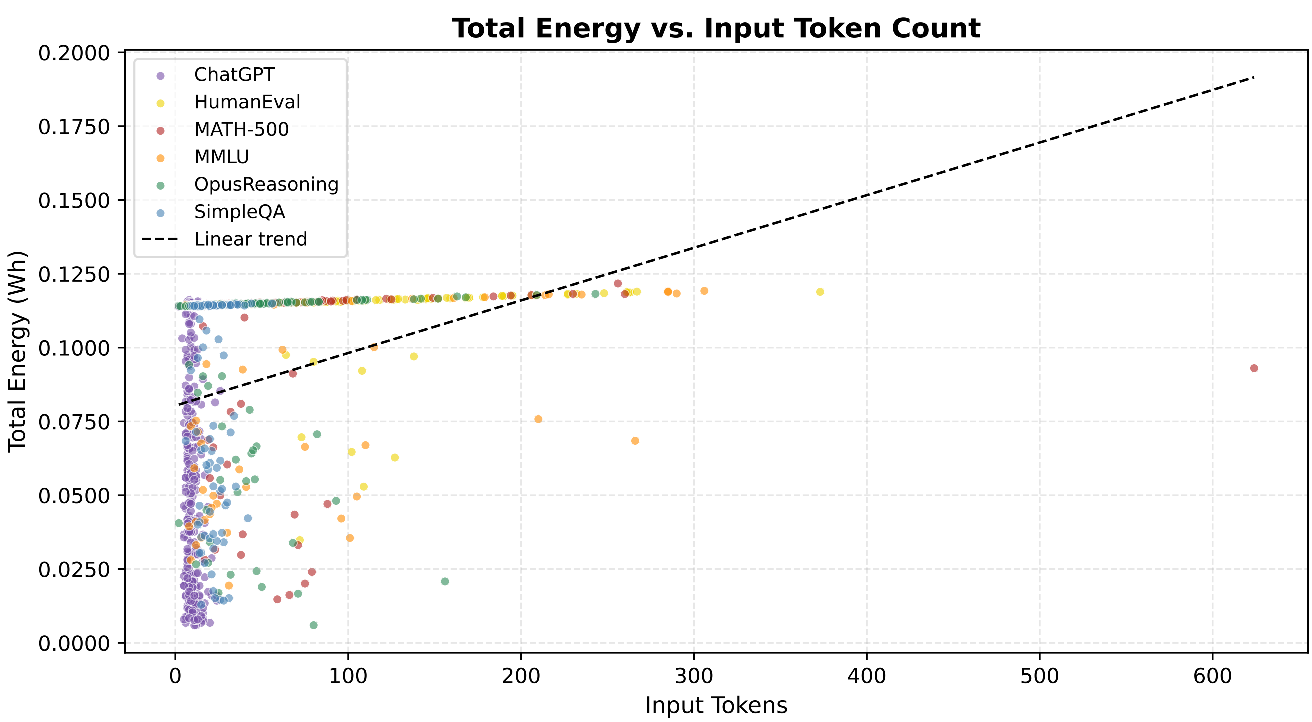

Figure 8: Distribution of Energy Cost per Prompt in Watt-Hours by Total Input Tokens.

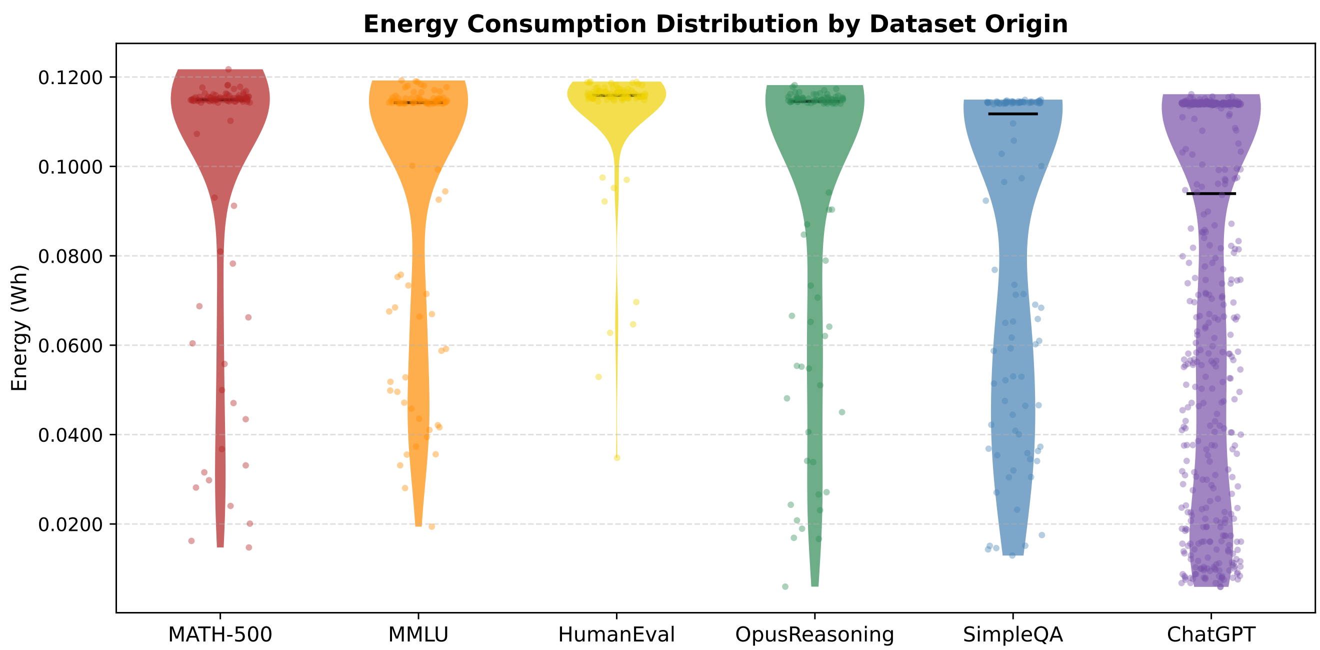

Figure 9: Distribution of Energy Costs per Prompt in Watt-Hours.

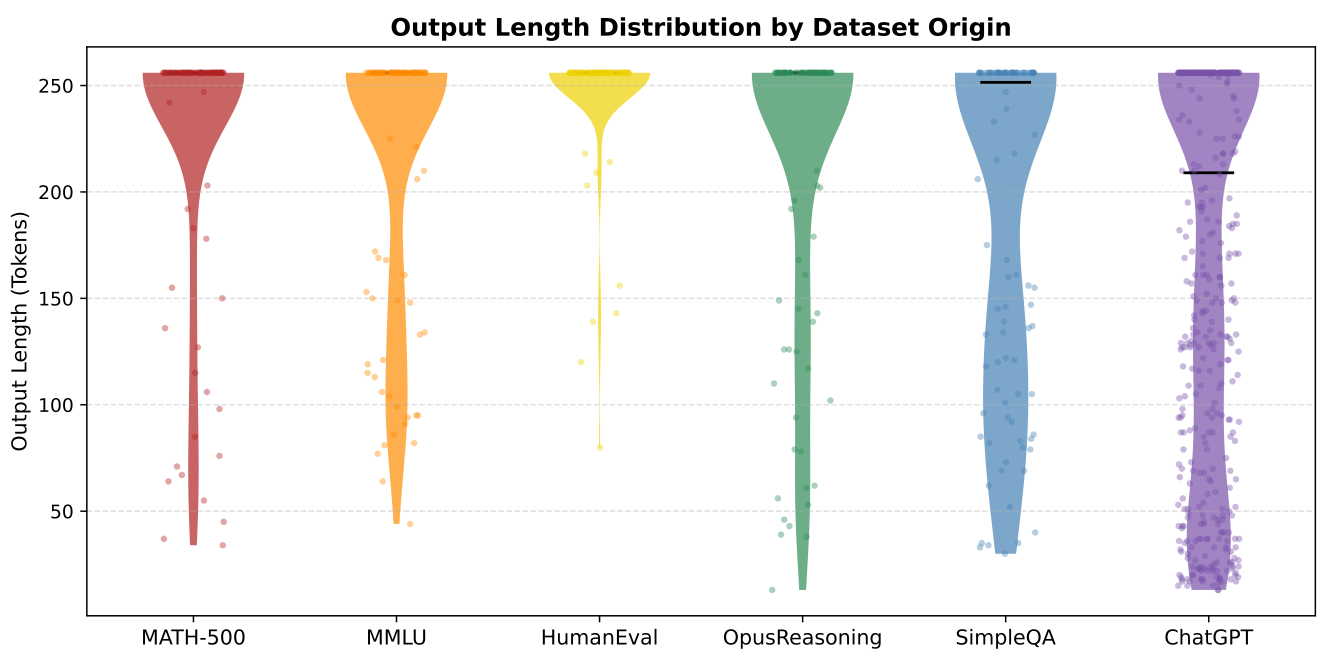

Figure 10: Distribution of Output Length per Prompt in Words (Tokens).

model architectures and hardware configurations. These works fall into three categories in the context of our study: experimental inference studies, simulated inference studies, and experimental training studies.

Experimental Inference Studies: These studies measure LLM energy consumption during inference under a range of hardware and software configurations. One notable contribution introduces TokenPowerBench, a benchmarking framework that evaluates LLM

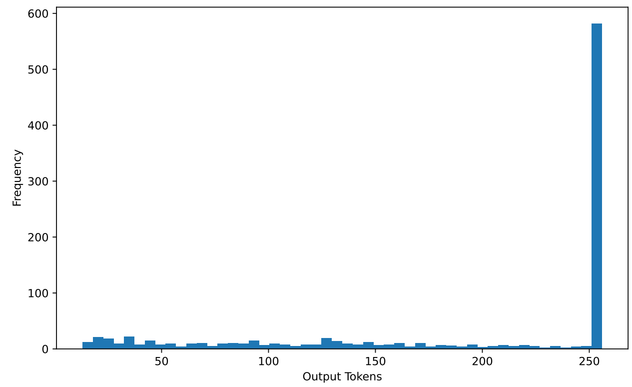

Figure 11: Distribution of Output Length for GPT-2.

performance through energy consumption per token [12]. A number of studies in this category quantify inference energy consumption in great detail, though their findings are largely oriented toward readers with a strong background in machine learning, leaving a gap our study can hopefully help fill [8][3][17][11].

Simulated Inference Studies: In this category, researchers have the same question but take a complementary approach. They estimate inference energy costs through mathematical modeling rather than direct measurement. These studies typically derive energy figures by calculating Joules from floating-point operation (FLOP) counts, or by instrumenting a simulated inference environment [22].

Experimental Training Studies: These constitute the most extensively developed area of this broader research topic. A notable study by Strubell et al. [20] established an early framework for quantifying the energy cost of training large language models, and subsequent work has expanded on these findings across varying hardware and sofrware configurations [7][18]. While direct comparisons across studies are complicated by differences in hardware, software, and experimental design, this body of work collectively provides important context for understanding the full energy footprint of a model's life-cycle.

Related studies also supported our choice of GPT model for experimentation [6] and confirmed the relationship between energy cost and output length [15].

Despite the range of existing research, relatively little attention has been paid to the energy cost of a single prompt. As LLMs are increasingly deployed at scale, understanding the cost of a singular interaction will be important to model realistic energy consumption. Furthermore, given the rapid pace of advancement in AI systems, continued investigation in this area is necessary to ensure that energy cost estimates remain current and account for improvements in model efficiency and hardware performance.

6 Conclusion

For lightweight models like GPT-2-124M and opt-125M, we found the average energy cost of a single prompt to be 0.97 Wh. This result is supported by both our experimental methodology and simulated calculations, and it aligns with the existing literature.

We found that output length had a significant impact on the energy cost of prompting an LLM. Prompts that produced over 100 tokens took significantly more energy than shorter responses.

While there are valid concerns about the long-term impact of AI on the environment and human intelligence, the cost of a single prompt to a lightweight model is comparable to a Google search. The real impact comes from training large state-of-the-art models; therefore the responsibility for AI's environmental impact falls to the companies training these models, and not to the consumer.

7 Future Work

We leave additional model testing and an expansion of the prompt dataset for future work. Particularly, we would like to address the energy costs for larger models with more parameters and for newly emerging model architectures.

Additional future work includes recording the content of model output to observe the impact of output quality on energy cost.

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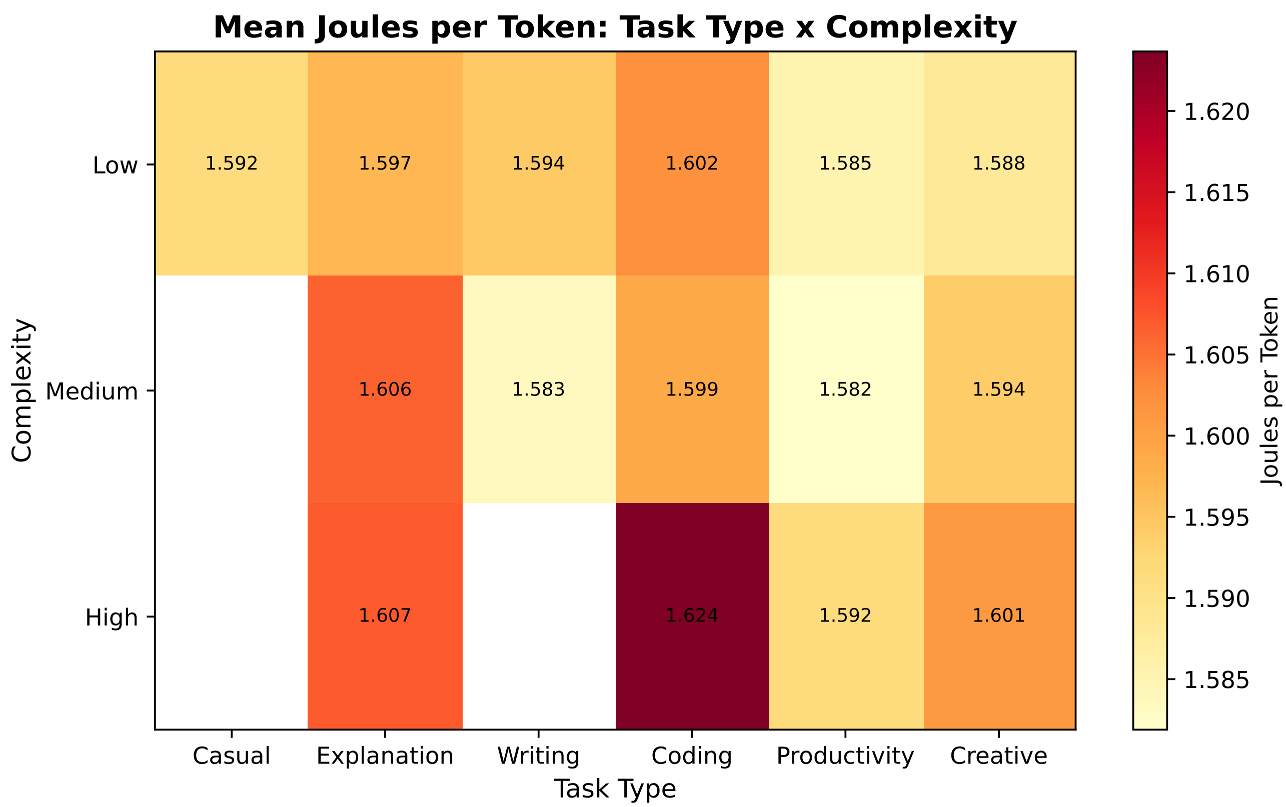

Figure 12: The amount of energy, in Joules, per input token, organized by task type and complexity. White boxes had no instances: for example, there were no high-complexity casual prompts.

Table 3: Model Hyperparameters

Parameter	Value
Max Tokens	250
No Repeat N-Gram Size	3

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A Model Parameters

Table 3 summarizes the hyperparameters used for all models. The purpose of setting maximum repeated n-gram size to three is to discourage our models from producing repetitive output and we cap both input and output token counts at 250 to control computational cost.

Received —; revised —; accepted —